Derivation of Cubic Splines: [adapted from Chapra]

Equation for Cubic Splines: $f_i = a_i + b_i(x - x_i) + c_i(x - x_i)^2 + d_i(x - x_i)^3$

Now, Let us consider 4 points then there will be 3 splines and 2knots.

Thus, for **n points**, there will be **(n-1) splines** and **(n-2) knots**.

Number of unknowns therefore is = 4(n-1) ; [4 for a_i , b_i , c_i , d_i]

Hence, We need to have 4(n-1) equations to solve for the unknowns. In the following context an example of 3 splines and then it is generalized for n points.

Equations for the splines may be written as follows:

For n=4; $S_1(x) = a_1 + b_1(x - x_1) + c_1(x - x_1)^2 + d_1(x - x_1)^3$

$$S_2(x) = a_2 + b_2(x - x_2) + c_2(x - x_2)^2 + d_2(x - x_2)^3$$

$$S_3(x) = a_3 + b_3(x - x_3) + c_3(x - x_3)^2 + d_3(x - x_3)^3$$

General : $S_i(x) = a_i + b_i(x - x_i) + c_i(x - x_i)^2 + d_i(x - x_i)^3$ **(1)**

<u>First Condition</u> : Splines pass through all the data points.

For n=4;

S_1 passes through (x_1, y_1): $S_1(x_1) = a_1 + b_1(x_1 - x_1) + c_1(x_1 - x_1)^2 + d_1(x_1 - x_1)^3$

or, $S_1(x_1) = a_1 = f_1$

S_2 passes through (x_2, y_2) : $S_2(x_2) = a_2 = f_2$

S_3 passes through (x_3, y_3) : $S_3(x_3) = a_3 = f_3$

General : $S_i(x_i) = a_i = f_i$ **(2)**

Again,

S_3 passes through (x_4, y_4) : $S_3(x_4) = a_3 + b_3(x_4 - x_3) + c_3(x_4 - x_3)^2 + d_3(x_4 - x_3)^3$

Ie., S_{n-1} passes through (x_n, y_n):

General: $S_{n-1}(x_n) = a_{n-1} + b_{n-1}(x_n - x_{n-1}) + c_{n-1}(x_n - x_{n-1})^2 + d_{n-1}(x_n - x_{n-1})^3$ **(3)**

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<u>Second Condition</u> : Second Derivative continuity : Second derivatives at the knots are equal.

Equation (1) can be differentiated twice to give : $S_i'(x) = b_i + 2c_i(x_{i+1} - x_i) + 3d_i(x_{i+1} - x_i)^2$

$$S_i''(x) = 2c_i + 6d_i(x_{i+1} - x_i)$$

or, $$S_i''(x) = 2c_i + 6d_i h_i \qquad \textbf{(4)}$$

For n=4 : $$S_1''(x)|_{(x_2,y_2)} = S_2''(x)|_{(x_2,y_2)}$$

$$2c_1 + 6d_1(x_2 - x_1) = 2c_2 + 6d_2(x_2 - x_2)$$

$$2c_1 + 6d_1 h_1 = 2c_2$$

$$S_2''(x)|_{(x_3,y_3)} = S_3''(x)|_{(x_3,y_3)}$$

$$2c_2 + 6d_2 h_2 = 2c_3$$

General : $$S_i''(x)|_{(x_{i+1},y_{i+1})} = S_{i+1}''(x)|_{(x_{i+1},y_{i+1})}$$

$$2c_i + 6d_i h_i = 2c_{i+1}$$

$$c_i + 3d_i h_i = c_{i+1} \qquad \textbf{(5)}$$

or, $$d_i = \frac{c_{i+1} - c_i}{3h_i} \qquad \textbf{(6)}$$

<u>Third Condition</u>: First derivative continuity : First derivatives at the knots are equal.

Equation (1) can be differentiated to give : $S_i'(x) = b_i + 2c_i(x_{i+1} - x_i) + 3d_i(x_{i+1} - x_i)^2$ **(7)**

or, $$S_i'(x) = b_i + 2c_i h_i + 3d_i h_i^2$$

For n=4 : $$S_1'(x)|_{(x_2,y_2)} = S_2'(x)|_{(x_2,y_2)}$$

$$b_1 + 2c_1(x_2 - x_1) + 3d_1(x_2 - x_1)^2 = b_2 + 2c_2(x_2 - x_2) + 3d_2(x_2 - x_2)^2$$

$$b_1 + 2c_1 h_1 + 3d_1 h_1^2 = b_2$$

$$S_2'(x)|_{(x_3,y_3)} = S_3'(x)|_{(x_3,y_3)}$$

$$b_2 + 2c_2 h_2 + 3d_2 h_2^2 = b_3$$

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General : $$S_i'(x)|_{(x_{i+1},y_{i+1})} = S_{i+1}'(x)|_{(x_{i+1},y_{i+1})}$$

$$b_i + 2c_i h_i + 3d_i h_i^2 = b_{i+1} \qquad \textbf{(8)}$$

equation (6) can be substituted in equation (8) to get :

$$b_{i+1} = b_i + h_i\left(c_i + c_{i+1}\right) \qquad \textbf{(9)}$$

Reducing the index of the equation by 1 we get :

$$b_i = b_{i-1} + h_{i-1}\left(c_{i-1} + c_i\right)$$

$$b_i - b_{i-1} = h_{i-1}\left(c_{i-1} + c_i\right) \qquad \textbf{(10)}$$

<u>Fourth Condition</u> : Continuity : Splines are continuous at the knots, ie.,

For n=4 ; $$S_1(x)|_{(x_2,y_2)} = S_2(x)|_{(x_2,y_2)}$$

$$a_1 + b_1\left(x_2 - x_1\right) + c_1\left(x_2 - x_1\right)^2 + d_1\left(x_2 - x_1\right)^3 = a_2 + b_2\left(x_2 - x_2\right) + c_2\left(x_2 - x_2\right)^2 + d_2\left(x_2 - x_2\right)^3$$

$$a_1 + b_1\left(x_2 - x_1\right) + c_1\left(x_2 - x_1\right)^2 + d_1\left(x_2 - x_1\right)^3 = a_2$$

$$S_2(x)|_{(x_3,y_3)} = S_3(x)|_{(x_3,y_3)}$$

$$a_2 + b_2\left(x_3 - x_2\right) + c_2\left(x_3 - x_2\right)^2 + d_2\left(x_3 - x_2\right)^3 = a_3 + b_3\left(x_3 - x_3\right) + c_3\left(x_3 - x_3\right)^2 + d_3\left(x_3 - x_3\right)^3$$

$$a_2 + b_2\left(x_3 - x_2\right) + c_2\left(x_3 - x_2\right)^2 + d_2\left(x_3 - x_2\right)^3 = a_3$$

General : $$S_i(x)|_{(x_{i+1},y_{i+1})} = S_{i+1}(x)|_{(x_{i+1},y_{i+1})}$$

$$a_i + b_i\left(x_{i+1} - x_i\right) + c_i\left(x_{i+1} - x_i\right)^2 + d_i\left(x_{i+1} - x_i\right)^3 = a_{i+1}$$

From Equation (2) we may write,

$$f_{i+1} = f_i + b_i\left(x_{i+1} - x_i\right) + c_i\left(x_{i+1} - x_i\right)^2 + d_i\left(x_{i+1} - x_i\right)^3$$

Now, $h_i = x_{i+1} - x_i$;

Therefore , $$f_{i+1} = f_i + b_i h_i + c_i h_i^2 + d_i h_i^3 \qquad \textbf{(11)}$$

equation (6) can be substituted in equation (11) to get :

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$$f_{i+1} = f_i + b_i h_i + c_i h_i^2 + \left(\frac{c_{i+1} - c_i}{3h_i} \right) h_i^3$$

$$f_{i+1} = f_i + b_i h_i + \frac{h_i^2}{3}\left(c_{i+1} + 2c_i\right)$$

$$b_i = \frac{f_{i+1} - f_i}{h_i} - \frac{h_i}{3}\left(c_{i+1} + 2c_i\right) \qquad \textbf{(12)}$$

Reducing the index of the equation by 1 we get :

$$b_{i-1} = \frac{f_i - f_{i-1}}{h_{i-1}} - \frac{h_{i-1}}{3}\left(c_i + 2c_{i-1}\right) \qquad \textbf{(13)}$$

Substracting equation (13) from (12) we get ;

$$b_i - b_{i-1} = \left[\frac{f_{i+1} - f_i}{h_i} - \frac{h_i}{3}\left(c_{i+1} + 2c_i\right) \right] - \left[\frac{f_i - f_{i-1}}{h_{i-1}} - \frac{h_{i-1}}{3}\left(c_i + 2c_{i-1}\right) \right] \qquad \textbf{(14)}$$

Equating equations (11) and (14) we get :

$$h_{i-1}\left(c_{i-1} + c_i\right) = \left[\frac{f_{i+1} - f_i}{h_i} - \frac{h_i}{3}\left(c_{i+1} + 2c_i\right) \right] - \left[\frac{f_i - f_{i-1}}{h_{i-1}} - \frac{h_{i-1}}{3}\left(c_i + 2c_{i-1}\right) \right]$$

$$\frac{h_{i-1}}{3}\left(c_i + 2c_{i-1}\right) - h_{i-1}\left(c_{i-1} + c_i\right) + \frac{h_i}{3}\left(c_{i+1} + 2c_i\right) = \frac{f_i - f_{i-1}}{h_{i-1}} - \frac{f_{i+1} - f_i}{h_i}$$

$$h_{i-1}c_{i-1} + 2\left(h_{i-1} + h_i\right)c_i + h_i c_{i+1} = 3\frac{f_{i+1} - f_i}{h_i} - 3\frac{f_i - f_{i-1}}{h_{i-1}} \qquad \textbf{(15)}$$

The equation can be made a little more concise by recognizing that the terms on the right hand side of equation (15) are finite differences, $f\left[x_i, x_j\right] = \dfrac{f_i - f_j}{x_i - x_j}$

Therefore, equation (15) may be re-written

$$\boxed{h_{i-1}c_{i-1} + 2\left(h_{i-1} + h_i\right)c_i + h_i c_{i+1} = 3\left(f\left[x_{i+1}, x_i\right] - f\left[x_i, x_{i-1}\right] \right)} \qquad \textbf{(16)}$$

Equation (15) can be written for the Knots, i=2,3,……,(n-1),which results in (n-2) simultaneous tridiagonal equations with n unknown coefficients, c_1, c_2, …… , c_n . Therefore if we now have two additional conditions, we can solve for the c's.

Once this is done, b's can be determined from equation (12) : $\boxed{b_i = \dfrac{f_{i+1} - f_i}{h_i} - \dfrac{h_i}{3}\left(c_{i+1} + 2c_i\right)}$

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And d 's can be determined from equation (6) : $\boxed{d_i = \frac{c_{i+1} - c_i}{3h_i}}$

<u>Cubic splines and End Conditions</u> :

1. **Natural Condition :** In this condition, the second derivative at the first and the last node is equal to zero.

From equation (1):

$$S_i(x) = a_i + b_i(x - x_i) + c_i(x - x_i)^2 + d_i(x - x_i)^3$$

$$S_i'(x) = b_i + 2c_i(x - x_i) + 3d_i(x - x_i)^2$$

$$S_i''(x) = 2c_i + 6d_i(x - x_i)$$

For the first node : $S_1''(x_1) = 0 = 2c_1 + 6d_1(x_1 - x_1)$ **(17)**

Therefore , $\boxed{c_1 = 0}$ **(18)**

For the last node : $S_{n-1}''(x_n) = 2c_{n-1} + 6d_{n-1}(x_n - x_{n-1}) = 0$

$$c_{n-1} + 3d_{n-1}h_{n-1} = 0$$

We can conveniently define an extraneous parameter c_n , in which equation (18) becomes

$$c_{n-1} + 6d_{n-1}h_{n-1} = c_n = 0$$

$$\boxed{c_n = 0} \quad \textbf{(19)}$$

For n points : equation (16) can be written as :

$$h_1c_1 + 2(h_1 + h_2)c_1 + h_2c_2 = 3\left(f[x_3, x_2] - f[x_2, x_1]\right)$$

$$h_2c_2 + 2(h_2 + h_3)c_2 + h_3c_3 = 3\left(f[x_4, x_3] - f[x_3, x_2]\right)$$

$$h_{n-2}c_{n-2} + 2(h_{n-2} + h_{n-1})c_{n-1} + h_{n-1}c_n = 3\left(f[x_n, x_{n-1}] - f[x_{n-1}, x_{n-2}]\right)$$

Hence the final equations may be written in matrix form as :

For n=4 points,

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ h_1 & 2(h_1 + h_2) & h_2 & 0 \\ 0 & h_2 & 2(h_2 + h_3) & h_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 3\left(f[x_3, x_2] - f[x_2, x_1]\right) \\ 3\left(f[x_4, x_3] - f[x_3, x_2]\right) \\ 0 \end{Bmatrix}$$

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For n points,

$$\begin{bmatrix} 1 & 0 & \cdots & \cdots & \cdots & \cdots & 0 \\ h_1 & 2(h_1+h_2) & h_2 & 0 & 0 & 0 & \vdots \\ 0 & h_2 & 2(h_2+h_3) & h_3 & 0 & 0 & \vdots \\ \vdots & 0 & \ddots & \ddots & \ddots & 0 & \vdots \\ \vdots & 0 & 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & 0 & 0 & 0 & h_{n-2} & 2(h_{n-2}+h_{n-1}) & h_{n-1} \\ 0 & \cdots & \cdots & \cdots & \cdots & 0 & 1 \end{bmatrix} \begin{Bmatrix} c_1 \\ c_2 \\ \vdots \\ \vdots \\ \vdots \\ c_{n-1} \\ c_n \end{Bmatrix} = \begin{Bmatrix} 0 \\ 3\left(f[x_3,x_2]-f[x_2,x_1]\right) \\ 3\left(f[x_4,x_3]-f[x_3,x_2]\right) \\ \vdots \\ \vdots \\ 3\left(f[x_n,x_{n-1}]-f[x_{n-1},x_{n-2}]\right) \\ 0 \end{Bmatrix}$$

As shown, the system is tridiagonal and hence efficient to solve.

2. **Clamped Condition :** in this condition f_1' and f_n' are specified at the first and last node. Hence,

   From equation (1) : $S_i(x) = a_i + b_i(x-x_i) + c_i(x-x_i)^2 + d_i(x-x_i)^3$

   $S_i'(x) = b_i + 2c_i(x-x_i) + 3d_i(x-x_i)^2$

   For the first node : $S_1'(x) = b_1 + 2c_1(x-x_1) + 3d_1(x-x_1)^2$

   $S_1'(x)|_{(x_1,y_1)} = f_1' = b_1 + 2c_1(x_1-x_1) + 3d_1(x_1-x_1)^2$

   $S_1'(x)|_{(x_1,y_1)} = f_1' = b_1$

   Substituting the above equation in equation (12): $b_i = \dfrac{f_{i+1}-f_i}{h_i} - \dfrac{h_i}{3}(c_{i+1}+2c_i)$, we get ,

   $$f_1' = \frac{f_2-f_1}{h_1} - \frac{h_1}{3}(2c_2+c_1)$$

   $$\frac{f_2-f_1}{h_1} - f_1' = \frac{h_1}{3}(c_2+2c_1)$$

   $$\boxed{c_2h_1 + 2h_1c_1 = 3f[x_2,x_1] - 3f_1'}$$

   For Last node : $S_{n-1}'(x) = b_{n-1} + 2c_{n-1}(x-x_{n-1}) + 3d_{n-1}(x-x_{n-1})^2$

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$$S_{n-1}'(x)|_{(x_n,y_n)} = f_n' = b_{n-1} + 2c_{n-1}(x_n - x_{n-1}) + 3d_{n-1}(x_n - x_{n-1})^2$$

$$S_{n-1}'(x)|_{(x_n,y_n)} = f_n' = b_{n-1} + 2c_{n-1}h_{n-1} + 3d_{n-1}h_{n-1}^{\ 2}$$

Again from equation (8)we know, $b_i + 2c_i h_i + 3d_i h_i^2 = b_{i+1}$

Hence, $\boxed{f_n' = b_n}$

From equation (9) we know , $b_{i+1} = b_i + h_i(c_i + c_{i+1})$

$$b_n = b_{n-1} + h_{n-1}(c_{n-1} + c_n)$$

$$f_n' = b_{n-1} + h_{n-1}(c_{n-1} + c_n)$$

From equation (13) we know, $b_{i-1} = \dfrac{f_i - f_{i-1}}{h_{i-1}} - \dfrac{h_{i-1}}{3}(c_i + 2c_{i-1})$

$$b_{n-1} = \frac{f_n - f_{n-1}}{h_{n-1}} - \frac{h_{n-1}}{3}(c_n + 2c_{n-1})$$

$$f_n' - h_{n-1}(c_{n-1} + c_n) = \frac{f_n - f_{n-1}}{h_{n-1}} - \frac{h_{n-1}}{3}(c_n + 2c_{n-1})$$

$$f_n' - h_{n-1}(c_{n-1} + c_n) = f[x_n, x_{n-1}] - \frac{h_{n-1}}{3}(c_n + 2c_{n-1})$$

$$\boxed{2h_{n-1}c_{n-1} + h_{n-1}c_n = 3\left(f_n' - f[x_n, x_{n-1}]\right)}$$

The final equations can now be written in a matrix form as :

$$\begin{bmatrix} 2h_1 & h_1 & 0 & \cdots & \cdots & \cdots & 0 \\ h_1 & 2(h_1+h_2) & h_2 & 0 & 0 & 0 & \vdots \\ 0 & h_2 & 2(h_2+h_3) & h_3 & 0 & 0 & \vdots \\ \vdots & 0 & \ddots & \ddots & \ddots & 0 & \vdots \\ \vdots & 0 & 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & 0 & 0 & 0 & h_{n-2} & 2(h_{n-2}+h_{n-1}) & h_{n-1} \\ 0 & \cdots & \cdots & \cdots & \cdots & 2h_{n-1} & h_{n-1} \end{bmatrix} \begin{Bmatrix} c_1 \\ c_2 \\ \vdots \\ \vdots \\ \vdots \\ c_{n-1} \\ c_n \end{Bmatrix} = \begin{Bmatrix} 3f[x_2, x_1] - 3f_1' \\ 3\left(f[x_3, x_2] - f[x_2, x_1]\right) \\ 3\left(f[x_4, x_3] - f[x_3, x_2]\right) \\ \vdots \\ \vdots \\ 3\left(f[x_n, x_{n-1}] - f[x_{n-1}, x_{n-2}]\right) \\ 3\left(f_n' - f[x_n, x_{n-1}]\right) \end{Bmatrix}$$

3. **Not –a- knot condition :** The third derivative at the second and next to last knots are continuous, therefore,

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$$S_i(x) = a_i + b_i(x - x_i) + c_i(x - x_i)^2 + d_i(x - x_i)^3$$

$$S_i''(x) = 2c_i + 6d_i(x_{i+1} - x_i)$$

$$S_i'''(x) = 6d_i$$

$$S_1'''(x)|_{(x_2, y_2)} = S_2'''(x)|_{(x_2, y_2)}$$

$$6d_1 = 6d_2 \qquad \text{or, } d_1 = d_2$$

From equation (6) we know , $\quad d_i = \dfrac{c_{i+1} - c_i}{3h_i}$

$$\frac{c_2 - c_1}{3h_1} = \frac{c_3 - c_2}{3h_2}$$

$$h_2c_2 - h_2c_1 = h_1c_3 - h_1c_2$$

$$\boxed{h_2c_1 - (h_1 + h_2)c_2 + h_1c_3 = 0}$$

For last but one node : $\quad S_{n-2}'''(x)|_{(x_{n-1}, y_{n-1})} = S_{n-1}'''(x)|_{(x_{n-1}, y_{n-1})}$

$$d_{n-1} = d_{n-2}$$

$$\frac{c_{n-1} - c_{n-2}}{3h_{n-2}} = \frac{c_n - c_{n-1}}{3h_{n-1}}$$

$$h_{n-1}c_{n-1} - h_{n-1}c_{n-2} = h_{n-2}c_n - h_{n-2}c_{n-1}$$

$$\boxed{h_{n-1}c_{n-2} - (h_{n-2} + h_{n-1})c_{n-1} + h_{n-2}c_n = 0}$$

$$\begin{bmatrix} h_2 & -(h_1+h_2) & h_1 & 0 & \cdots & \cdots & 0 \\ h_1 & 2(h_1+h_2) & h_2 & 0 & 0 & 0 & \vdots \\ 0 & h_2 & 2(h_2+h_3) & h_3 & 0 & 0 & \vdots \\ \vdots & 0 & \ddots & \ddots & \ddots & 0 & \vdots \\ \vdots & 0 & 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & 0 & 0 & 0 & h_{n-2} & 2(h_{n-2}+h_{n-1}) & h_{n-1} \\ 0 & \cdots & \cdots & \cdots & h_{n-1} & -(h_{n-2}+h_{n-1}) & h_{n-2} \end{bmatrix} \begin{Bmatrix} c_1 \\ c_2 \\ \vdots \\ \vdots \\ \vdots \\ c_{n-1} \\ c_n \end{Bmatrix} = \begin{Bmatrix} 0 \\ 3\left(f[x_3, x_2] - f[x_2, x_1]\right) \\ 3\left(f[x_4, x_3] - f[x_3, x_2]\right) \\ \vdots \\ \vdots \\ 3\left(f[x_n, x_{n-1}] - f[x_{n-1}, x_{n-2}]\right) \\ 0 \end{Bmatrix}$$

Thus for the above mentioned conditions the Splines may be determined.

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GENERAL LINEAR LEAST SQUARES (adapted from Mathews & Finks)

Simple linear, polynomial and multi linear all three belongs to the following general linear least squares model:

$$y = a_0 z^0 + a_1 z^1 + a_2 z^2 + a_3 z^3 + a_4 z^4 + a_5 z^5 . + ...a_m z^m + e \tag{1}$$

It can easily be seen how simple linear and multi linear regression fall within this model ie $z^0 = 1, z^1 = x_1, z^2 = x_2, \cdots, z^m = x_m$. Polynomial regression is also included if the basis functions are $z^0 = 1, z^1 = x^1, z^2 = x^2, \cdots, z^m = x^m$

'n' number of data points $(Z_k \quad y_k)$ and $z^0, z^1, z^2, z^3, z^4, z^5, \cdots, z^m$ are $(m+1)$ independent functions. So for $k \quad th$ deta point the equation is:

$$y_k = a_0 z_k^0 + a_1 z_k^1 + a_2 z_k^2 + a_3 z_k^3 + a_4 z_k^4 + a_5 z_k^5 + ...a_m z_k^m + e_k$$

IF coefficient vector is {a} then

$$\begin{aligned} \{y\} &= [Z]\{a\} + \{e\} \\ \left[[Z]' \quad [Z]\right] \quad \{\mathrm{a}\} &= [Z]'\{y\} + [Z]'\{e\} \end{aligned} \tag{2}$$

Where

$$\begin{aligned} [Z] &= \begin{bmatrix} Z_1^0 & Z_1^1 & Z_1^2 & \cdots & Z_1^m \\ Z_2^0 & Z_2^1 & Z_2^2 & \cdots & Z_2^m \\ Z_3^0 & Z_3^1 & Z_3^2 & \ldots & Z_3^m \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ Z_n^0 & Z_n^1 & Z_n^2 & \ldots & Z_n^m \end{bmatrix} \\ \{a\} &= \{a_0 \quad a_1 \quad a_2 \quad \cdots \quad a_m\}' \\ \{y\} &= \{y_1 \quad y_2 \quad y_3 \quad \cdots \quad y_n\}' \\ \{e\} &= \{e_1 \quad e_2 \quad e_3 \quad \cdots \quad e_n\}' \end{aligned} \tag{3}$$

If residual part e become zero then $\{e\} = \{0 \quad 0 \quad \cdots \quad 0\}'$

$$\left[[Z]' \quad [Z]\right]\{a\} = [Z]'\{y\} \tag{4}$$

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$$\left[[Z]' \quad [Z]\right] = \begin{bmatrix} Z_1^0 & Z_2^0 & Z_3^0 & \cdots & Z_n^0 \\ Z_1^1 & Z_2^1 & Z_3^1 & \cdots & Z_n^1 \\ Z_1^2 & Z_2^2 & Z_3^2 & \cdots & Z_n^2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ Z_1^m & Z_2^m & Z_3^m & \cdots & Z_n^m \end{bmatrix} \begin{bmatrix} Z_1^0 & Z_1^1 & Z_1^2 & \cdots & Z_1^m \\ Z_2^0 & Z_2^1 & Z_2^2 & \cdots & Z_2^m \\ Z_3^0 & Z_3^1 & Z_3^2 & \cdots & Z_3^m \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ Z_n^0 & Z_n^1 & Z_n^2 & \cdots & Z_n^m \end{bmatrix}$$

$$= \begin{bmatrix} \sum_{k=1}^{N} Z_k^0 Z_k^0 & \sum_{k=1}^{n} Z_k^0 Z_k^1 & \sum_{k=1}^{n} Z_k^0 Z_k^2 & \cdots & \sum_{k=1}^{n} Z_k^0 Z_k^m \\ \sum_{k=1}^{n} Z_k^1 Z_k^0 & \sum_{k=1}^{n} Z_k^1 Z_k^1 & \sum_{k=1}^{n} Z_k^1 Z_k^2 & \cdots & \sum_{k=1}^{n} Z_k^1 Z_k^m \\ \sum_{k=1}^{n} Z_k^2 Z_k^0 & \sum_{k=1}^{n} Z_k^2 Z_k^1 & \sum_{k=1}^{n} Z_k^2 Z_k^2 & \cdots & \sum_{k=1}^{n} Z_k^2 Z_k^m \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \sum_{k=1}^{n} Z_k^m Z_k^0 & \sum_{k=1}^{n} Z_k^m Z_k^1 & \sum_{k=1}^{n} Z_k^m Z_k^2 & \cdots & \sum_{k=1}^{n} Z_k^m Z_k^m \end{bmatrix} \tag{5}$$

The element of $i \quad th$ row $j \quad th$ column of $\left[[Z'] \quad [Z]\right]$ is =

$$\sum_{k}^{n} Z_k^{i-1} Z_k^{j-1} = Z_1^{i-1} Z_1^{j-1} + Z_2^{i-1} Z_2^{j-1} + Z_3^{i-1} Z_3^{j-1} + Z_4^{i-1} Z_4^{j-1} + \cdots\cdots\cdots\cdots + Z_n^{i-1} Z_n^{j-1}$$

$$\left\{[Z]' \quad \{y\}\right\} = \begin{bmatrix} Z_1^0 & Z_2^0 & Z_3^0 & \cdots & Z_n^0 \\ Z_1^1 & Z_2^1 & Z_3^1 & \cdots & Z_n^1 \\ Z_1^2 & Z_2^2 & Z_3^2 & \cdots & Z_n^2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ Z_1^m & Z_2^m & Z_3^m & \cdots & Z_n^m \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{bmatrix}$$

$$= \begin{Bmatrix} \sum_{k=1}^{n} Z_k^0 y_k \\ \sum_{k=1}^{n} Z_k^1 y_k \\ \sum_{k=1}^{n} Z_k^2 y_k \\ \vdots \\ \sum_{k=1}^{n} Z_k^m y_k \end{Bmatrix} \tag{6}$$

Final expression is:

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$$\begin{bmatrix} \sum_{k=1}^{N} Z_k^0 Z_k^0 & \sum_{k=1}^{n} Z_k^0 Z_k^1 & \sum_{k=1}^{n} Z_k^0 Z_k^2 & \cdots & \sum_{k=1}^{n} Z_k^0 Z_k^m \\ \sum_{k=1}^{n} Z_k^1 Z_k^0 & \sum_{k=1}^{n} Z_k^1 Z_k^1 & \sum_{k=1}^{n} Z_k^1 Z_k^2 & \cdots & \sum_{k=1}^{n} Z_k^1 Z_k^m \\ \sum_{k=1}^{n} Z_k^2 Z_k^0 & \sum_{k=1}^{n} Z_k^2 Z_k^1 & \sum_{k=1}^{n} Z_k^2 Z_k^2 & \cdots & \sum_{k=1}^{n} Z_k^2 Z_k^m \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \sum_{k=1}^{n} Z_k^m Z_k^0 & \sum_{k=1}^{n} Z_k^m Z_k^1 & \sum_{k=1}^{n} Z_k^m Z_k^2 & \cdots & \sum_{k=1}^{n} Z_k^m Z_k^m \end{bmatrix} \begin{Bmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_m \end{Bmatrix} = \begin{Bmatrix} \sum_{k=1}^{n} Z_k^0 y_k \\ \sum_{k=1}^{n} Z_k^1 y_k \\ \sum_{k=1}^{n} Z_k^2 y_k \\ \vdots \\ \sum_{k=1}^{n} Z_k^m y_k \end{Bmatrix} \tag{7}$$

After solving $\{a\} = \{a_0 \quad a_1 \quad a_2 \quad \cdots \quad a_m\}'$ the model equation is:

$$y = a_0 z^0 + a_1 z^1 + a_2 z^2 + a_3 z^3 + a_4 z^4 + a_5 z^5 . + ...a_m z^m \tag{8}$$

y = prediction of least square fit

From equation (1) e is residual, and then square of residual is Sr

$$s_r = \sum_{k=1}^{n} \left[y_k - \left(a_0 z_k^0 + a_1 z_k^1 + a_2 z_k^2 + a_3 z_k^3 + a_4 z_k^4 + a_5 z_k^5 . + ...a_m z_k^m \right) \right]^2 \tag{9}$$

To minimize Sr it is necessary that each partial derivative be zero and these results in the system of equations

$$\frac{\partial Sr}{\partial a_k} = 0 \tag{10}$$

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$$
\begin{aligned}
\frac{\partial s_r}{\partial a_0} &= \sum_{k=1}^{n} -2z_k^0\left[y_k - \left(a_0 z_k^0 + a_1 z_k^1 + a_2 z_k^2 + a_3 z_k^3 + a_4 z_k^4 + a_5 z_k^5 + ...a_m z_k^m\right)\right] = 0 \\
\frac{\partial s_r}{\partial a_1} &= \sum_{k=1}^{n} -2z_k^1\left[y_k - \left(a_0 z_k^0 + a_1 z_k^1 + a_2 z_k^2 + a_3 z_k^3 + a_4 z_i^4 + a_5 z_k^5 + ...a_m z_k^m\right)\right] = 0 \\
\frac{\partial s_r}{\partial a_2} &= \sum_{k=1}^{n} -2z_2^0\left[y_k - \left(a_0 z_k^0 + a_1 z_k^1 + a_2 z_k^2 + a_3 z_k^3 + a_4 z_k^4 + a_5 z_k^5 + ...a_m z_k^m\right)\right] = 0 \\
\frac{\partial s_r}{\partial a_3} &= \sum_{k=1}^{n} -2z_k^3\left[y_k - \left(a_0 z_k^0 + a_1 z_k^1 + a_2 z_k^2 + a_3 z_k^3 + a_4 z_k^4 + a_5 z_k^5 + ...a_m z_k^m\right)\right] = 0 \\
&\vdots \\
&\vdots \\
&\vdots \\
\frac{\partial s_r}{\partial a_m} &= \sum_{k=1}^{n} -2z_k^m\left[y_k - \left(a_0 z_k^0 + a_1 z_k^1 + a_2 z_k^2 + a_3 z_k^3 + a_4 z_k^4 + a_5 z_k^5 + ...a_m z_k^m\right)\right] = 0
\end{aligned}
\tag{11}
$$

Rearranging equation (11) wrt to $a_0 \quad a_1 \quad a_2 \quad \cdots \quad a_m$

$$
\begin{aligned}
a_0\sum_{k=1}^{n}\left(z_k^0\right)^2 + a_1\sum_{k=1}^{n} z_k^0 z_k^1 + a_2\sum_{k=1}^{n} z_k^0 z_k^2 + a_3\sum_{k=1}^{n} z_k^0 z_k^3 + ... + a_m\sum_{k=1}^{n} z_k^0 z_k^m &= \sum_{k=1}^{n} Z_k^0 y_k \\
a_0\sum_{k=1}^{n} z_k^1 z_k^0 + a_1\sum_{k=1}^{n}\left(z_k^1\right)^2 + a_2\sum_{k=1}^{n} z_k^1 z_k^2 + a_3\sum_{k=1}^{n} z_k^1 z_k^3 + ... + a_m\sum_{k=1}^{n} z_k^1 z_k^m &= \sum_{k=1}^{n} Z_k^1 y_k \\
a_0\sum_{k=1}^{n} z_i^2 z_i^0 + a_1\sum_{k=1}^{n} z_k^2 z_k^1 + a_2\sum_{k=1}^{n}\left(z_k^2\right)^2 + a_3\sum_{k=1}^{n} z_k^2 z_k^3 + ... + a_m\sum_{k=1}^{n} z_k^1 z_k^m &= \sum_{k=1}^{n} Z_k^2 y_k \\
&\vdots \\
&\vdots \\
&\vdots \\
a_0\sum_{k=1}^{n} z_i^m z_i^0 + a_1\sum_{k=1}^{n} z_k^m z_k^1 + a_2\sum_{k=1}^{n} z_k^m z_k^2 + a_3\sum_{k=1}^{n} z_k^m z_k^3 + ... + a_m\sum_{k=1}^{n}\left(z_k^m\right)^2 &= \sum_{k=1}^{n} Z_k^m y_k
\end{aligned}
\tag{12}
$$

It can be easily recognized as a system of m+1 linear equations m+1 unknown.

So

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$$[A]\{a\}=\{B\}$$

$$or,$$

$$\begin{bmatrix} \sum_{k=1}^{N} Z_k^0 Z_k^0 & \sum_{k=1}^{n} Z_k^0 Z_k^1 & \sum_{k=1}^{n} Z_k^0 Z_k^2 & \cdots & \sum_{k=1}^{n} Z_k^0 Z_k^m \\ \sum_{k=1}^{n} Z_k^1 Z_k^0 & \sum_{k=1}^{n} Z_k^1 Z_k^1 & \sum_{k=1}^{n} Z_k^1 Z_k^2 & \cdots & \sum_{k=1}^{n} Z_k^1 Z_k^m \\ \sum_{k=1}^{n} Z_k^2 Z_k^0 & \sum_{k=1}^{n} Z_k^2 Z_k^1 & \sum_{k=1}^{n} Z_k^2 Z_k^2 & \cdots & \sum_{k=1}^{n} Z_k^2 Z_k^m \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \sum_{k=1}^{n} Z_k^m Z_k^0 & \sum_{k=1}^{n} Z_k^m Z_k^1 & \sum_{k=1}^{n} Z_k^m Z_k^2 & \cdots & \sum_{k=1}^{n} Z_k^m Z_k^m \end{bmatrix} \begin{Bmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_m \end{Bmatrix} = \begin{Bmatrix} \sum_{k=1}^{n} Z_k^0 y_k \\ \sum_{k=1}^{n} Z_k^1 y_k \\ \sum_{k=1}^{n} Z_k^2 y_k \\ \vdots \\ \sum_{k=1}^{n} Z_k^m y_k \end{Bmatrix} \tag{13}$$

Out cum of the process from equation (8) to (13) is

$$\left[[Z'] \quad [Z]\right]\{a\}=[Z']\{y\}$$

So, previous assumption in equation (4) was correct. After solving $\{a\}$, Sr can be calculated as

$$\begin{aligned} s_r &= \sum_{k=1}^{n}\left[y_k-\left(a_0 z_k^0+a_1 z_k^1+a_2 z_k^2+a_3 z_k^3+a_4 z_k^4+a_5 z_k^5.+.....a_m z_k^m\right)\right]^2 \\ &= \sum_{k=1}^{n}\left[y_k-y_k\right] \end{aligned} \tag{14}$$

$$S_t=\sum_{k=1}^{n}\left(y_k \quad - \quad \bar{y}\right)^2$$

$$r^2=\frac{S_t \quad - \quad S_r}{S_t}=1-\frac{S_r}{S_t}=1-\frac{\sum_{k=1}^{n}\left(y_k \quad - \quad y\right)^2}{\sum_{k=1}^{n}\left(y_k \quad - \quad \bar{y}\right)^2} \tag{15}$$

y = prediction of least square method

$\bar{y}$ = mean of y_k

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Quadratic Spline:

The objective in quadratic ***splines*** is to derive a 2nd order polynomial for each interval between data points. The polynomial for each interval can be generally represented as: $S_i(\mathrm{x}_i) = a_i + b_i(\mathrm{x}-\mathrm{x}_i) + c_i(\mathrm{x}-\mathrm{x}_i)^2 \cdots\cdots(1)$

For n data points $(\mathrm{i} = 1, 2 \cdots n)$, there are $(n-1)$ intervals and, consequently, $3(n-1)$ unknown constants (the a's, b's & c's) to evaluate. So, $3(n-1)$ equations or conditions are required to evaluate the unknowns.

It can be developed as follows:

- **1:** The function must pass through all the points.

$$f_i = a_i + b_i(\mathrm{x}_i - \mathrm{x}_i) + c_i(\mathrm{x}_i - \mathrm{x}_i)^2$$

Or, $f_i = a_i \cdots\cdots(2)$

So the constant in each spline must be equal to the value of the dependent variable at the beginning of the interval. Result can be incorporated into $Eq(1)$

$$S_i(\mathrm{x}_i) = f_i + b_i(\mathrm{x}-\mathrm{x}_i) + c_i(\mathrm{x}-\mathrm{x}_i)^2$$

For $(n-1)^{th}$ spline must pass through (x_n, y_n) point. So,

$$f_{n-1} = a_{n-1} + b_{n-1}(\mathrm{x}_n - \mathrm{x}_{n-1}) + c_{n-1}(\mathrm{x}_n - \mathrm{x}_{n-1})^2 \cdots\cdots(3)$$

The equation can be simplified mathematically by defining the width of the ith interval as $h_i = (x_{i+1} - x_i)$

So $Eq(3)$ simplifies to

$$f_{n-1} = a_{n-1} + b_{n-1}\mathrm{h}_{n-1} + c_{n-1}\mathrm{h}_{n-1}{}^2$$

Number of conditions to be evaluated has now reduced to $[3(n-1)-(n-1)-1] = (2n-1)$

- **2:** The function values of adjacent polynomials must be equal at knots. This condition can be written for knot $(i+1)$ as

$$f_i + b_i(\mathrm{x}_{i+1} - \mathrm{x}_i) + c_i(\mathrm{x}_{i+1} - \mathrm{x}_i)^2 = f_{i+1} + b_{i+1}(\mathrm{x}_{i+1} - \mathrm{x}_{i+1}) + c_{i+1}(\mathrm{x}_{i+1} - \mathrm{x}_{i+1})^2 + c_i(\mathrm{x}_{i+1} - \mathrm{x}_{i+1})^2 \cdots(4)$$
$$f_i + b_i h_i + c_i h_i^2 = f_{i+1} \cdots\cdots(5)$$

The equation can be written for nodes $i = 1, 2, 3, \cdots(n-1)$. So there are $(n-1)$ conditions, Number of conditions to be evaluated has now reduced to $[(2n-1)-(n-1)] = n$

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- **3:** The first derivatives at the interior nodes must be equal. It means that adjacent splines will be joined smoothly. The $Eq(1)$ can be differentiated to yield

$$S_i^{'}(\mathrm{x}) = b_i + 2c_i(\mathrm{x} - \mathrm{x}_i)$$

The equation of the derivatives at interior node $(i+1)$ can therefore be written as

$$b_i + 2c_i h_i = b_{i+1} \cdots\cdots (6)$$

Writing this equation for all the interior nodes amounts to $(n-2)$ conditions. This means that there is 1 remaining condition.

- **4:** Assume the 2nd derivative is zero at the first point. 2nd derivative of the $Eq(1)$ is $2c_i$. So, from $Eq(6)$

$$c_1 = 0 \cdots\cdots\cdots (7)$$

Putting this condition in $Eq(5)$ for $i = 1$

$$b_1 h_1 = (f_2 - f_1) \cdots\cdots (8)$$

From $Eq(6)$ $\quad c_i = \dfrac{(b_{i+1} - b_i)}{2h_i} \cdots\cdots (9)$

Putting $Eq(8)$ in $Eq(5)$ $\quad b_i h_i + b_{i+1} h_i = 2(f_{i+1} - f_i) \cdots\cdots (10)$

So putting $i = 1, 2, \cdots, n-1$ in to $Eq(9)$ there are $(n-1)$ equations with (n) variable.

$$\begin{array}{rcl} b_1 h_1 + b_2 h_1 & = & 2(f_2 - f_1) \\ b_2 h_2 + b_3 h_2 & = & 2(f_3 - f_2) \\ \vdots & \vdots & \vdots \\ b_{n-1} h_{n-1} + b_n h_{n-1} & = & 2(f_n - f_{n-1}) \end{array} \cdots\cdots (11)$$

Here we are assuming b_n as an extra variable. We have $(n-1)$ splines so $(n-1)$ b are there.

From $Eq(8) \,\&\, Eq(11)$

$$b_1 = b_2 \quad b_1 = b_2$$

$$\begin{bmatrix} h_1 & 0 & \cdots & 0 \\ h_2 & h_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & 0 \\ 0 & \cdots & h_{n-1} & h_{n-1} \end{bmatrix} \begin{bmatrix} b_2 \\ b_3 \\ \vdots \\ b_n \end{bmatrix} = \begin{bmatrix} (f_2 - f_1) \\ 2(f_3 - f_2) \\ \vdots \\ 2(f_n - f_{n-1}) \end{bmatrix}$$

After solving $b_1, b_2 \ldots, b_n$ we can calculate values of $c_1, c_2 \cdots c_{n-1}$ from equation (9).

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Taylors Series

$$y_{i+1} = y_i + f(x_i, y_i)h + \frac{f'(x_i, y_i)}{2!}h^2 + \cdots$$

Now, $f'(x_i, y_i)$ can be written by chain-rule differentiation

$$f'(x_i, y_i) = \frac{\partial f(x,y)}{\partial x} + \frac{\partial f(x,y)}{\partial y}\frac{dy}{dx}$$

Using the above equation in L1

$$y_{i+1} = y_i + f(x_i, y_i)h + \left(\frac{\partial f(x,y)}{\partial x} + \frac{\partial f(x,y)}{\partial y}\frac{dy}{dx}\right)\frac{h^2}{2!} + \cdots$$

Using $\frac{dy}{dx} = f(x_i, y_i)$ in the above equation

$$y_{i+1} = y_i + f(x_i, y_i)h + \left(\frac{\partial f(x,y)}{\partial x} + \frac{\partial f(x,y)}{\partial y}f(x_i, y_i)\right)\frac{h^2}{2!} + \cdots$$

Second Order RK Method is given by

$$y_{i+1} = y_i + (a_1k_1 + a_2k_2)h$$

$k_1 = f(x_i, y_i)$ and $k_2 = f(x_i + p_1h,\ y_i + q_{11}k_1h)$

Taylors Series for a two variable function

$$g(x + r, y + s) = g(x, y) + r\frac{\partial g}{\partial x} + s\frac{\partial g}{\partial y} + \cdots$$

Using the above equation for k_2

$$f(x_i + p_1h,\ y_i + q_{11}k_1h) = f(x_i, y_i) + p_1h\frac{\partial f(x,y)}{\partial x} + q_{11}k_1h\frac{\partial f(x,y)}{\partial y} + \cdots$$

Using the above equation in L1

$$y_{i+1} = y_i + a_1hf(x_i, y_i) + a_2hf(x_i, y_i) + a_2p_1h^2\frac{\partial f}{\partial x} + a_2q_{11}h^2f(x_i, y_i)\frac{\partial f}{\partial y} + \cdots$$

Group like terms

$$y_{i+1} = y_i + (a_1 + a_2)f(x_i, y_i)h + [a_2p_1\frac{\partial f}{\partial x} + a_2q_{11}f(x_i, y_i)\frac{\partial f}{\partial y}]h^2 + \cdots$$

Comparing the above two equations, we get

$$(a_1 + a_2) = 1;$$
$$a_2p_1 = \frac{1}{2};$$
$$a_2q_{11} = \frac{1}{2}$$

Taken from Numerical Methods for Engineers – Chapra & Canale

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